Comparing Statistical Models and Neural Networks in Polar Motion X-Component Forecasting:

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Abstract:

Precise polar motion (PM) forecast is directly related to satellite navigation and climate studies (Jin et al., 2021). Traditional methodologies and machine learning (ML) have been poplar in modern climate modeling (Wang & Zhang, 2023; Yu et al., 2024). This study addressed both approaches in comparison, investigating Autoregressive Integrated Moving Average (ARIMA) and Artificial Neural Network's (ANN) performance in seasonal geophysical signal forecast .

We collected polar motion data from the International Earth Rotation and Reference Systems Service (IERS) and global temperature data from NASA GISS Global Land-Ocean Temperature Index (GLOTI). We applied Kolmogorov-Zurbenko (KZ) filter to isolate low-frequency trends. An ARIMA and an ANN were each implemented on training data for a 6-month signal prediction.

Residual and statistical analysis showed that the ARIMA achieved superior accuracy than the ANN by the Diebold-Mariano (DM) test. The ANN's recursive architecture has shown to over-interpret noise as signals, causing overfitting patterns and phase lags in forecasting. These results demonstrated that traditional methodologies can outperform neural networks in noncomplex, seasonal forecasts. This study can provide critical insights to space agencies and climate researchers, presenting comparative analysis on two popular methodologies and their corresponding performances in polar motion forecasting.

Background Information:

- Polar motion (PM) is the movement of Earth's rotation axis relative to its crust; it's a fundamental component of Earth Orientation Parameters (EOP)
- Polar motion has been critical for space navigation, geodetic systems, and climate studies (Jin et al., 2021; Wang et al., 2024).
- Global surface temperature has been a primary metric for anthropogenic climate change, directly influencing Earth's rotations. (Kong et al., 2020).
- Before 21st century, polar motion prediction relied on linear models and least-squares extrapolation (Yao et al., 2013).
- In the modern age, the Fourier-based approaches and machine learning have been introduced to capture Chandler and annual wobbles (Zhao & Lei, 2018).

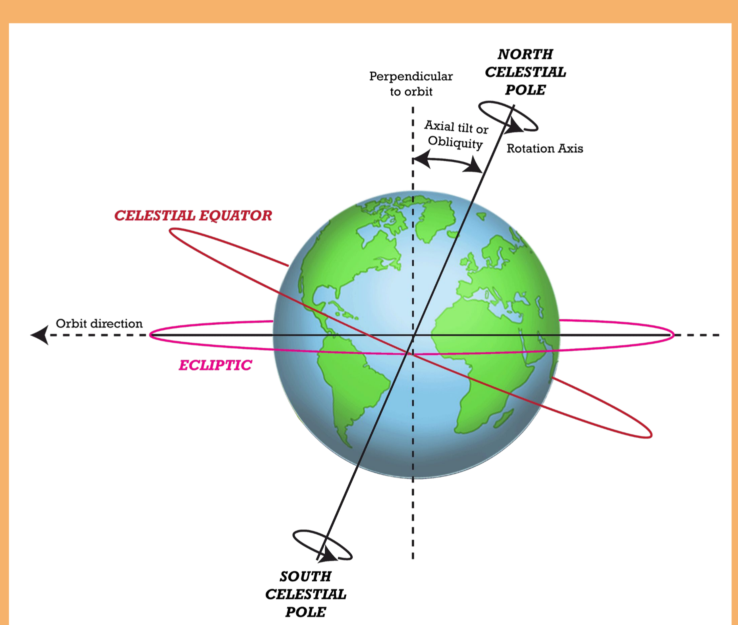

Figure 1: Conceptual figure showing Earth's orientation axis and polar motion.

Method/Materials:

- **Polar Motion Data**
  - Sourced from IERS (International Earth Rotation Service) (see Figure 2).
  - PM_X (polar motion x-component) in milliarcseconds (mas).
- **Global Temperature Data**
  - Sourced from NASA GISS GLOTI (Global Land-Ocean Temperature Index) (see Figure 2).
- **Kolmogorov-Zurbenko (KZ) filter**
  - Applied to PM_X to smooth out noise.
- **Time-Series Forecasting Models**
  - ARIMA (Autoregressive integrated moving average).
  - ANN (Artificial Neural Network)
- **RStudio (v.4.2.1) Key Packages**
  - Data: dplyr, data.table, zoo.
  - Visualization: ggplot2, patchwork.
  - Modeling: forecast, nnet, tseries.

Results:

$$Y_t = \frac{1}{m}\sum_{j=-k}^{k}(X_{t+j})$$

- KZ-Filter equation utilized for the moving average (Zurbenko, 1986).

$$(1-\sum_{i=1}^{p}\phi_i B^i)(1-B)^d X_t = c+(1+\sum_{j=1}^{q}\theta_j B^j)\epsilon_t+\sum_{k=1}^{2}\left[a_k\cos\left(\tfrac{2\pi kt}{365}\right)+b_k\sin\left(\tfrac{2\pi kt}{365}\right)\right]$$

- ARIMA + Fourier equation utilized to predict polar motion. It uses past values and yearly cycles

$$\hat{X}_t = \beta_0+\sum_{h=1}^{8} w_h\cdot g\left(\sum_{i=1}^{3} w_{hi}z_{ti}+b_h\right)$$

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- ANN equation utilized to learn patterns from past polar motion data. It recursively predicts future values for forecast (Wang et al., 2024).

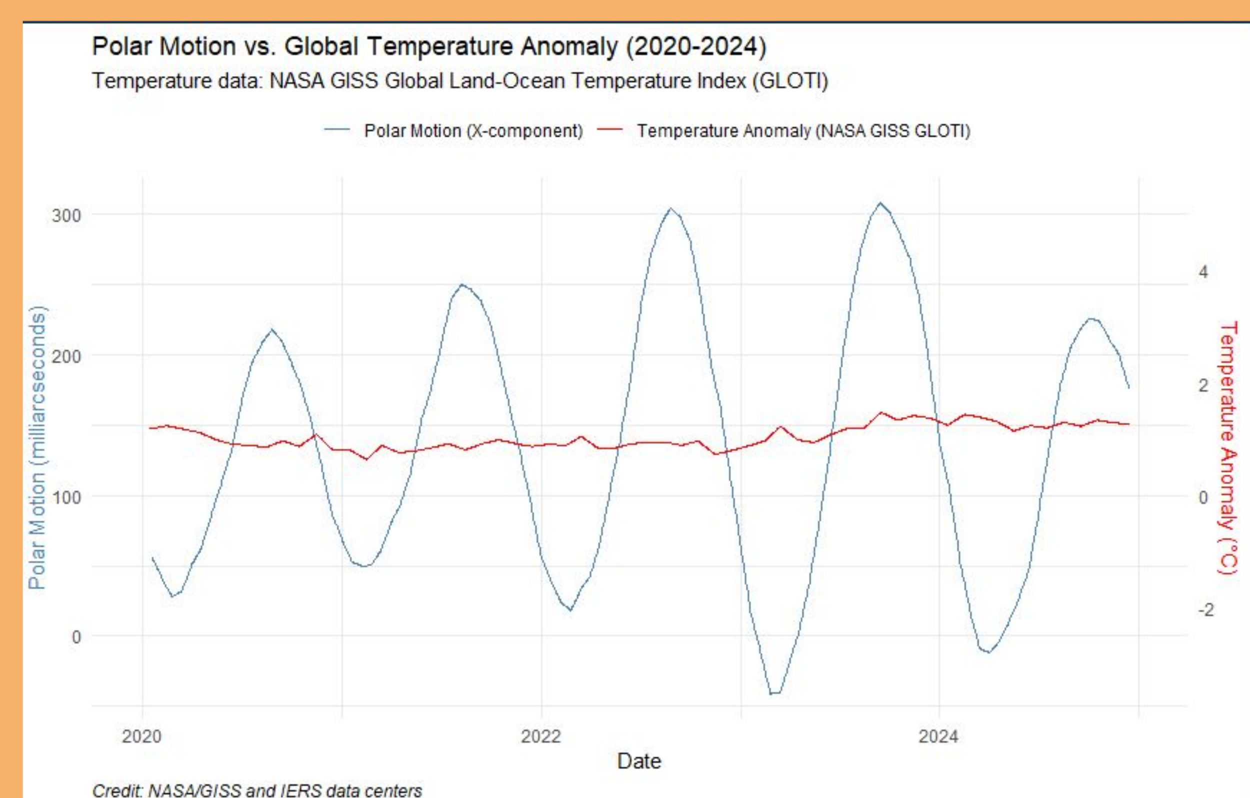

Figure 2. Polar Motion (X-component) vs. Global Temperature Anomaly (2020–2024) over time. Data sources: NASA/GISS and IERS.

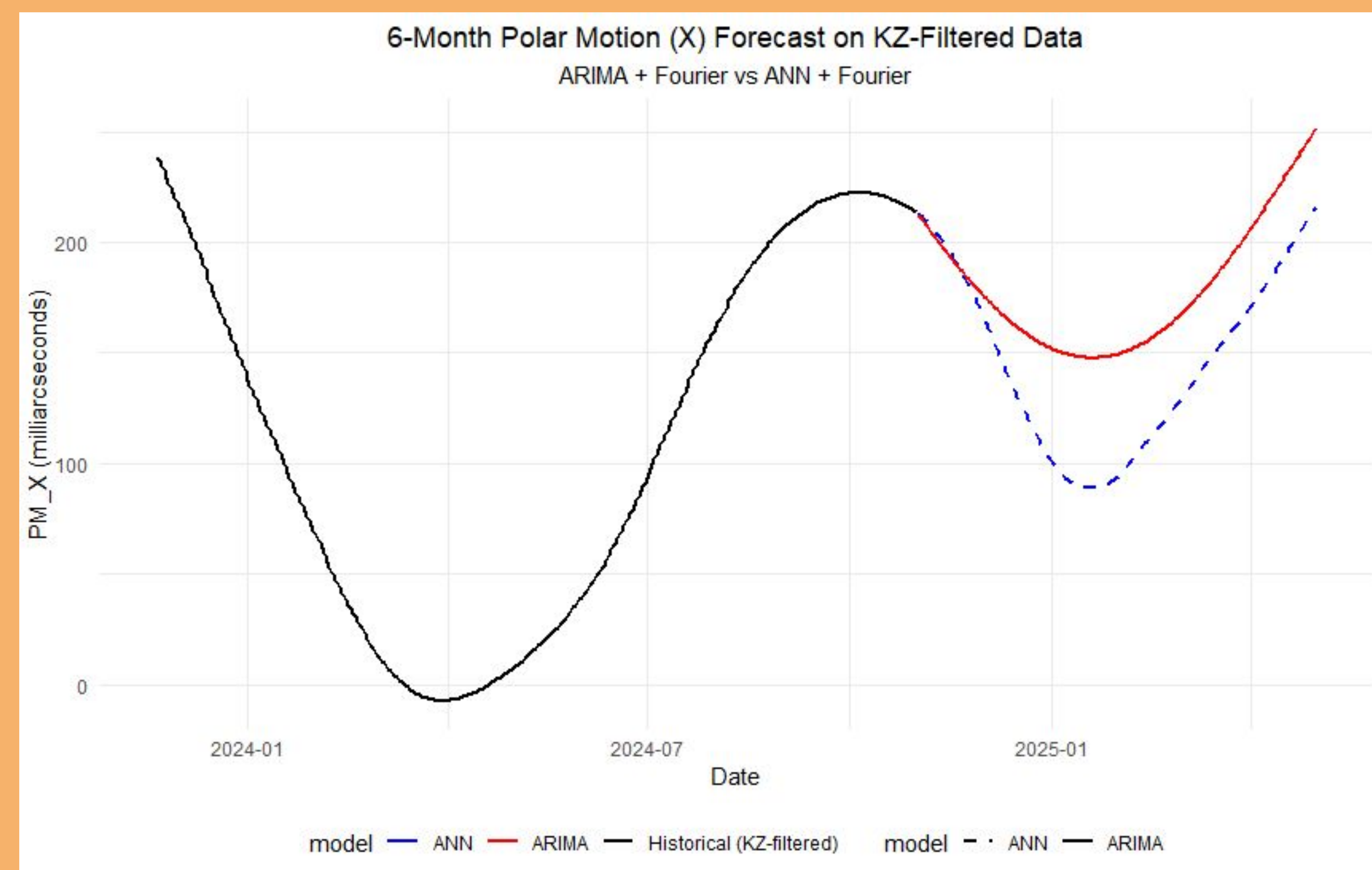

Figure 3: 6-month PM_X forecast comparison between ARIMA and ANN on KZ-filtered data. ARIMA closely follows the historical data's seasonal turning points, while ANN exhibits 1 to 2 month delay in capturing these reversals. ARIMA maintains a signal range similar to the historical data (125–225 mas), while ANN gives a more extreme swings (85–225 mas range compared to historical data's 110–210 mas range). ARIMA predicts a moderate downturn consistent with the KZ-filtered trend, while ANN projects an unrealistic sharp rebound.

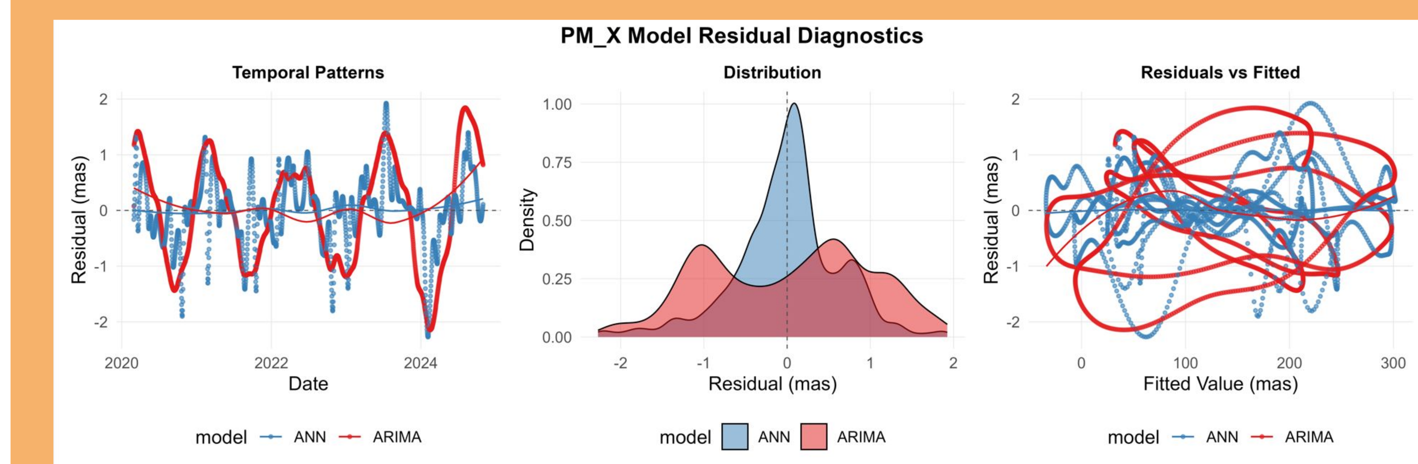

Figure 4: Comparison of ARIMA and ANN residuals. ARIMA's Temporal Residual scatters randomly around zero, while ANN's contains clusters of high and low values that are not random nor ideal. ARIMA's Residual Distribution shows a symmetric bell curve that matches closely with a normal curve, while ANN's is right-skewed with fat tails and contains too many extreme values. ARIMA's Residuals vs Fitted Values is evenly spread at all levels, while ANN's has spread that's wider at extremes.

	Residual Properties				Tests	
Model	Mean (mas)	SD (mas)	Skewness	Kurtosis	Jarque-Bera p	DM Test p
ARIMA + Fourier	0.32	12.4	0.15	2.9	0.120	NA
ANN	-1.45	18.7	0.82	4.1	0.003	0.03

Note:

SD: Standard Deviation; DM: Diebold-Mariano test comparing ANN to ARIMA

Table 1: Residual Statistics for PM_X Forecasting Models. ARIMA's skewness is ideally near zero, while ANN's skewness suggests greater positive errors and biased predictions. ARIMA's Kurtosis is close to three, resembling ideally to a normal distribution, while ANN's Kurtosis is much higher than three, indicating more outliers and worse predictions.

Discussion/Conclusion:

- ARIMA + Fourier model outperformed the ANN in forecasting polar motion (PM_X)
  - Lower residual SD (12.4 mas vs. 18.7 mas) (see Figure 4)
  - Better phase accuracy (see Figure 3)
  - Statistically superior forecasts (DM test: p = 0.03) (see Table 1)
- The ANN showed limitations in seasonal geophysical forecast
  - The ANN amplified errors due to its sensitivity to noise
  - ANN's heteroskedastic residuals suggest that it requires physical constraints (see Figure 4)
- The KZ filter effectively isolated low-frequency trends

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